# CS3 – Bone Cage for Spinal Implants: Exam Summary

### What was the design challenge?

The challenge was to design a spinal fusion cage that supports the spine at first, but gradually transfers load to new bone as it grows — using an adaptive modulus. The aim is to prevent stress shielding and promote long-term fusion.

### How does the cage ensure patient safety?

- Ti outer layer on the Apophseal ring (strong cortical bone) for stability.

- Mg inner layer degrades slowly to reduce stiffness over time (adaptive modulus).

- Chitosan coating on Mg slows degradation, preventing early collapse.

- Slot-in inserts (fast/medium/slow) allow customisation depending on patient bone quality.

### How does the design promote bone formation?

- Porous + lattice structure allows blood and nutrient flow.

- Concave pores mimic natural bone pits → osteoblasts love them.

- Nanoscale surface features increase protein adsorption, attract bone cells.

- HA coating (or nanoscale Ti) improves bone bonding.

- rhBMP-2 in centre chamber triggers new bone growth.

- Piezoelectric coatings (like PVDF) create signals that stimulate bone when patient moves.

### Why is radiolucency important and how is it achieved?

- Helps doctors see if bone is growing using X-rays/CT.

- Ti causes artefacts, so use Mg or PEEK inner parts (less radiodense).

- Design can be adaptive radiolucent: radiopaque early, radiolucent later as Mg degrades.

## Compare different material choices

| Material | Pros | Cons |
|---|---|---|
| Titanium (Ti) | Strong, good for screw fixation, easy to modify surface | Radiopaque, stress shielding |
| PEEK | Radiolucent, flexible | Poor cell attachment, hard to shape pores |
| Mg Alloy | Bioactive, degrades over time, radiolucent | Can degrade too fast (needs coating) |
| PLA | Biodegradable, proven | Acidic by-products, not ideal for bone growth |

## What are some added design features?

- Silver coating: Prevents infection, safe in low doses.

- Copper coating: Might help grow blood vessels.

- TPMS lattices: Reduce stiffness but keep strength.

- 3D printed using SLM or EBM, then surface-treated for better bone bonding.

## Bonus: A One-Line Summary (for viva or intro)

This spinal cage is a smart, layered implant with a strong Ti outer shell and a resorbable Mg inner core, designed to support spinal fusion safely while stimulating bone growth and allowing doctors to track healing over time.